MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
Discipline Specific Elective (DSE) –MG5DSECHE304 - MEDICINAL CHEMISTRY
MODEL QUESTION PAPER
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

1. Mention any two domains that fall under the scope of medicinal chemistry. (CO-1, U)
2. Define the terms pharmacology and toxicology (CO-1, K)
3. Analyze how the drug selectivity influences the numerical value of a therapeutic index. (CO-1, An)
4. Outline type I and type II drug receptors? (CO-2, U)
5. Define bioavailability of a drug and mention any two factors that can affect the bioavailability of a drug (CO-2, U)
6. A protein is associated with a disease but inhibition of the protein produces no therapeutic effect. What does this suggest about the target? (CO-1, An)
7. Explain the basic idea of the induced fit theory of drug action. (CO-2,U)
8. Outline the difference between cholinergic stimulants and cholinergic blockers. (CO-3, U)
9. Explain anaesthetics with an example. (CO-3, U)
10. Explain what is meant by an antimalarial drug, with one example (CO-3, U)
11. Explain the concept of gene therapy (CO-5, U)
12. Write a short note on sustained and controlled release drug formulations. (CO-5, U)

[10 × 2 = 20]

Part B

Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

13. Once a promising molecule is identified as a lead compound, it moves into the lead optimization phase. Distinguish between the goals of identifying a lead compound and optimizing a lead compound. (CO-1, An)
14. Glutathione exhibits less than 10% systemic bioavailability when administered orally. Suggest an alternative route of administration where it can achieve maximum bioavailability, providing reasons for your choice. (CO-1, A)
15. Categorize Prodrugs based on their site of bioactivation with suitable examples.

(CO-2, A)

16. A metabolite contains a suitable functional group for conjugation. Suggest a possible Phase II reaction and state its likely effect on excretion. (CO-2, A)
17. Define antibiotics and explain the use of chloramphenicol as an example. (CO-3, U)
18. Illustrate tranquilizers and summarize their uses. (CO-3, U)
19. Explain the concept and importance of pharmacovigilance in ensuring drug safety. (CO-4, U)
20. Briefly discuss the chemistry behind the application of liposomes as the drug delivery agents (CO-5, U)

[6 × 5 = 30]

Part C

Short Essay Questions
Answer any 2 questions
Each question carries 10 marks

21. You are the Chief Scientific Officer in charge of clinical trials of a novel drug for hypertension. Develop a brief plan for the clinical trials. (CO-1, A)
22. Explain adsorption, absorption, distribution, metabolism and elimination (ADME) of the drug in detail. (CO-2, U)
23. Compare the therapeutic outcomes of mimicking the adrenergic system versus inhibiting it. (CO-3, U)
24. Classify adverse drug reactions according to their type and severity. Explain each category with suitable examples. (CO-4, U)

[2 × 10 = 20]
